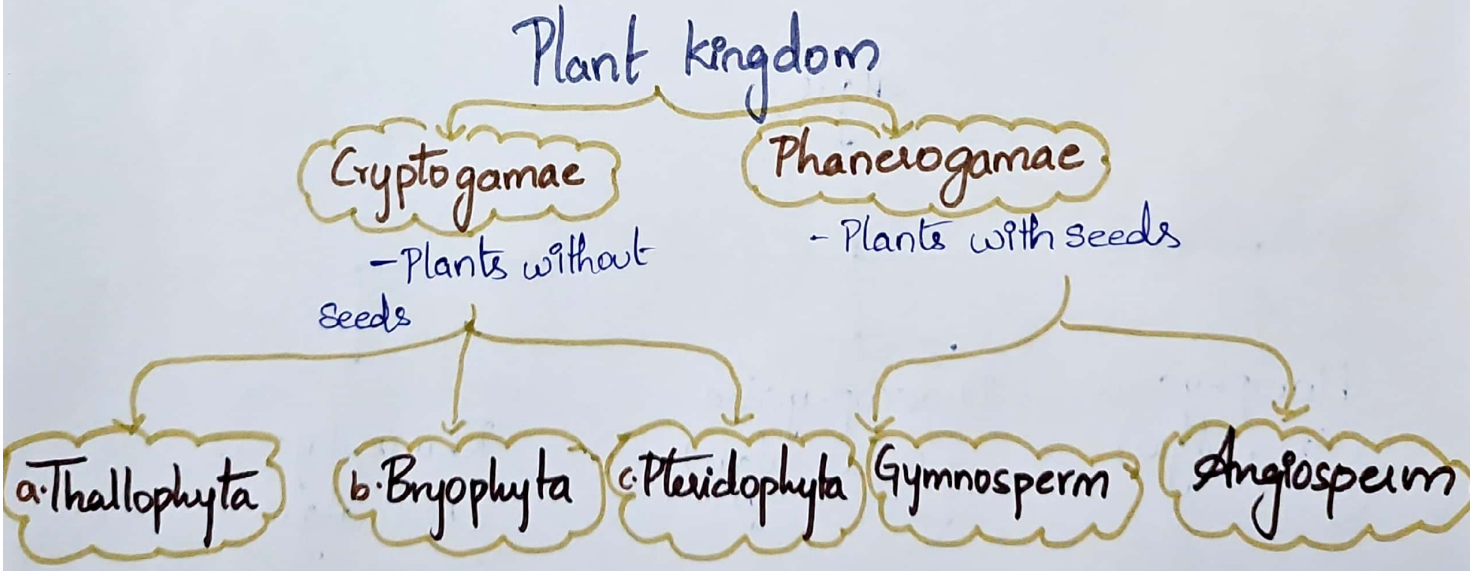
Plant Kingdom

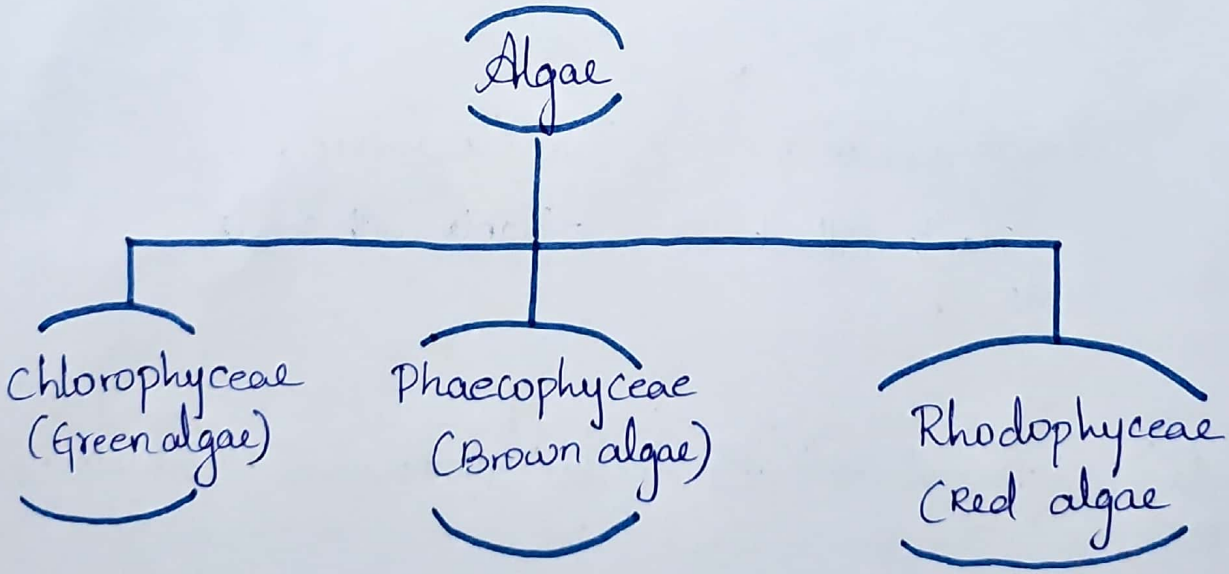
- Eukaryotic, multicellular, chlorophyll containing and having cell wall are grouped in the kingdom Plantae, popularly known as **Plant kingdom**
- Phylogenetic system of classification based on evolutionary relationship is presently used for classifying plants.
- **Numerical Taxonomy** use computer by assigning code for each character and analyzing the features.
- **Cytotaxonomy** is based on cytological information like chromosome number, structure and behaviour.
- **Chemotaxonomy** uses chemical constituents of plants to resolve the confusion.



a. **Thallophyta** - Comprises the simplest plants which possess undifferentiated (or) thallus like forms, reproductive organs single celled called gametangia. It includes only. **Algae.**

* Characteristics of Algae-

- Plant body is thallus, which may be unicellular, colonial filamentous (or) parenchymatous.
- Usually aquatic but a few are also found in moist terrestrial habitats like tree trunks, wet rocks, moist soil etc.
- Vascular tissues and mechanical tissues are absent.
- Reproduction is vegetative by fragmentation, asexual by spore formation [zoospores] and sexual reproduction by fusion of two gametes which may be Isogamous - **Spirogyra**, Anisogamous - **Chlamydomonas** (or) Oogamous - **Volvox**.
- Life cycle is various - haplontic, diplontic (or) diplohaplontic.



Economic Importance:

(1) A number of brown algae [Laminaria, Sargassum] are used as food in some countries.

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(ii) Fucus and Laminaria are rich source of Iodine.
(iii) Laminaria and Ascophyllum have antibiotic properties.
(iv) Alginic acid is obtained from fucus and Sargassum, which is used as emulsion.

Classes	Common Name	Major Pigments	stored food	cell wall	Flagellar Number	Habitat
Chlorophyceae	Green algae	Chlorophyll a, b	Starch	Cellulose	2-8 equal apical	Fresh water, salt wate Brackis wate
Phaeophyceae	Brown algae	Chlorophyll a, c fucoxanthin	Mannitol Laminarin	Cellulose + algin	2 unequal lateral	Fresh, Brackis and salt wate
Rhodophyceae	Red algae	Chlorophyll a, d Phycoerythrin	Floridean Starch	Cellulose Pectin + Poly sulphate ester	Absent	Fresh, Brackis salt water

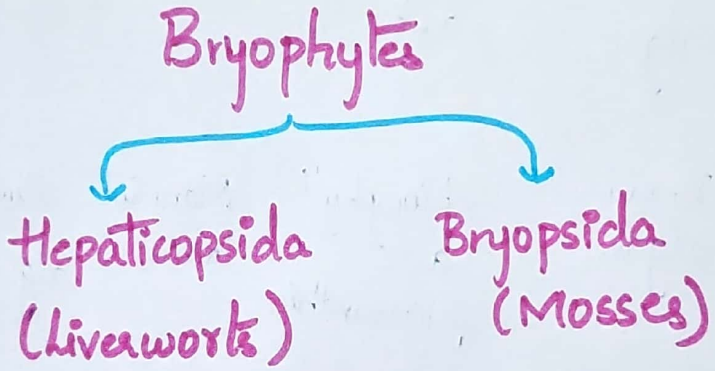
* **Bryophytes** – They are non-vascular mosses and liverworts that grow in moist shady region. They are called amphibians of plant kingdom because these plants live on soil but dependent on water for sexual reproduction.

Characteristic features –

- Live in damp and shady habitats, found to grow during rainy season on damp soil, rocks, walls etc.

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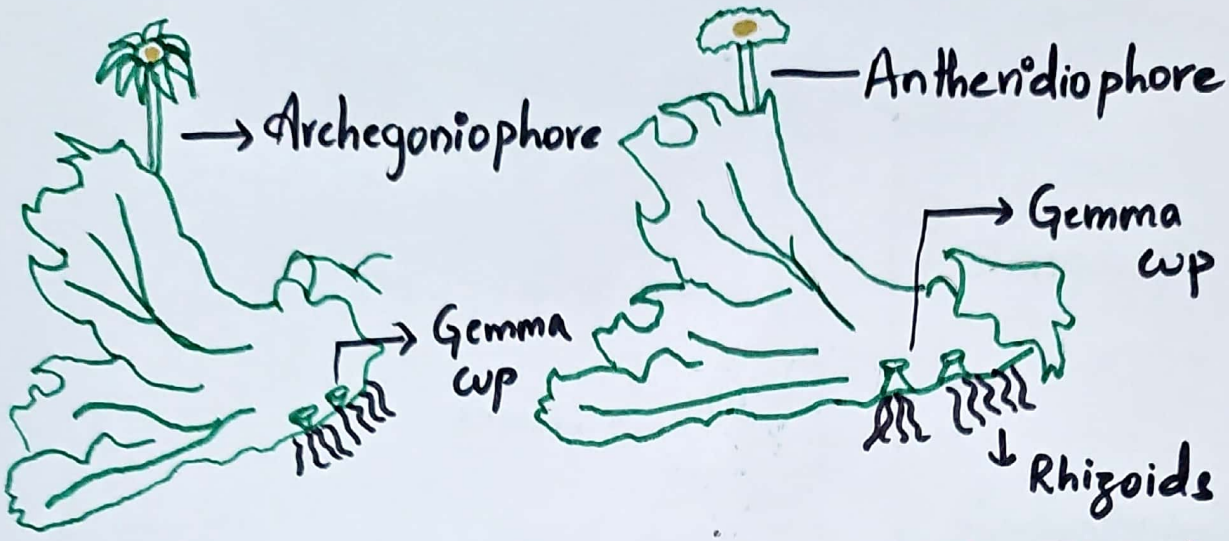
- The dominant phase (or) plant body is free living gametophyte
- Roots are absent but contain rhizoids.
- Vegetative reproduction is by fragmentation, tubers, gammae, buds etc. sex are multicellular and jacketed. Antheridium and archegonium produce male and female gametes called antherozoids and egg (or) oospore
- Sporophyte is parasite on gametophyte.



* Liverworts:

- The thallus is dorsoventrally flattened, dichotomously branched with or without leaf-like appendage.
- Unicellular rhizoids, multicellular scales and completely parasitic sporophyte (or) Sporangium.
- Asexual reproduction takes place by fragmentation thallus (or) formation of specialized structure called gemmae cups. The gemmae becomes detached from the parent body and germinate to form new individuals.
- During sexual reproduction male & female sex organs are produced on same thallus (or) different.
- The sporophyte is differentiated into foot, seta and capsule.

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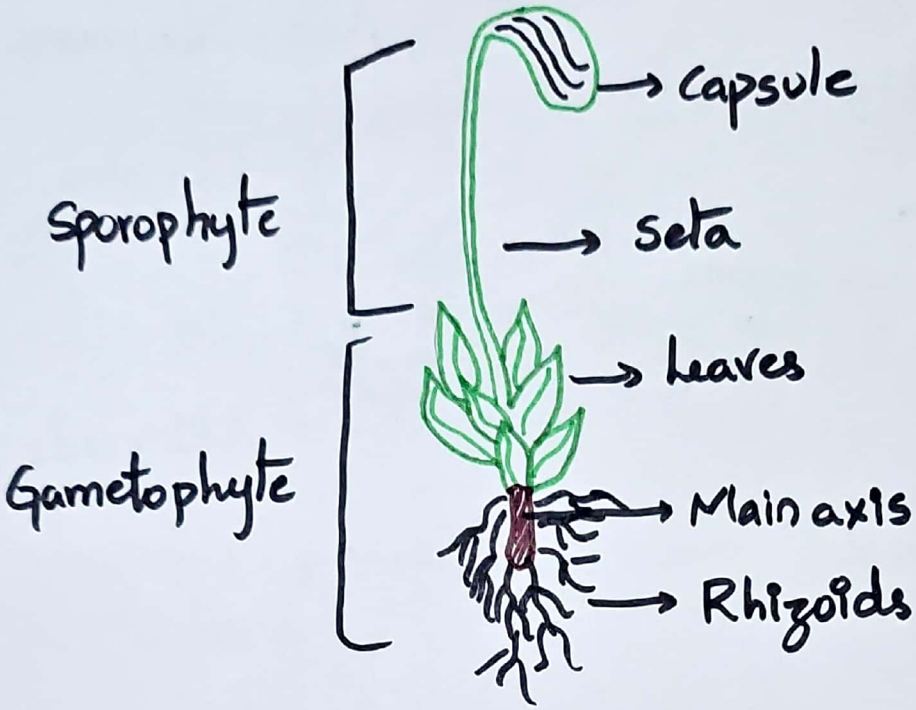
Marchantia male thallus

Marchantia female thallus.

* Mosses:

* The gametophyte of mosses consists of two-stages - the first stage is protonema stage, which develops directly from spores and creeping, green, frequently filamentous. The second stage is leafy stage which develops from secondary protonema as lateral bud having upright slender axes bearing spirally arranged leaves.
- Vegetative reproduction by the fragmentation and budding in secondary protonema. A sex organ develops on leafy shoots.
- Sporophytes in mosses are more developed and consist of foot, seta and capsule.
- Common examples are Funaria, Polytrichum, Sphagnum etc.

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* Pteridophytes :

- They are seedless vascular plants that have sporophytic plant body and inconspicuous gametophyte. Sporophytic plant body is differentiated into true stem, roots and leaves.
- Vascular tissues are present but vessels are absent from xylem and companion cells and sieve tube are absent.
- Sporophytes bear sporangia that are subtend by leaf like appendages called sporophylls. In some plants [selaginella] compact structure called strobili or cone is formed.
- Sporangia produce spores by meiosis in spore mother cells. Spores germinate to produce multicellular thalloid prothallus.
- Gametophyte bears male and female sex organ called

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antheridia and archegonia. Water is required for fertilisation of male and female gametes.

- Most of pteridophytes produces spores are of similar kind [**homosporous**] but in selginella and salvinia, spores are of two kinds [**heterosporous**] larger called megaspore that produce female gametophyte and smaller microspore that produce male gametes.

* Gymnosperms:

- Gymnosperms are those plants in which the ovules are not enclosed inside the ovary wall and remain exposed before and after fertilisation.
- They are perennial and woody, forming either bushes or trees. Some are very large (Sequoia sempervirens) and others are very small (Zamia pygmia).
- Stem may be unbranched (cycas) or branched (Pinus). Root is tap. Leaves may be simple or compound.
- They are heterosporous, produce haploid microspore and megaspore in male and female strobili respectively.
- Male and female gametophytes do not have independent free-living existence. Pollination occurs through air and zygote develops into embryo and ovules into seeds.
- Examples - Pines, Cycas, Cedrus, Ginkgo etc.

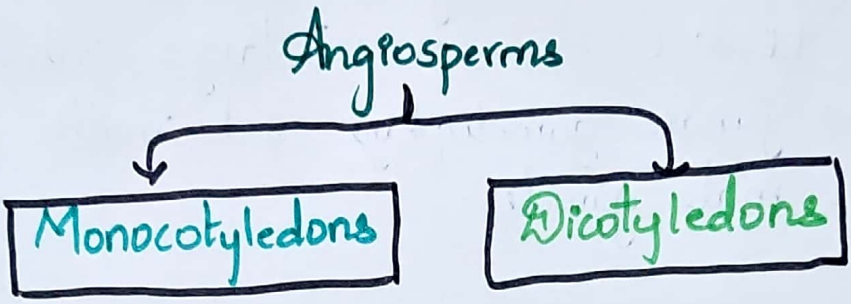
* Angiosperms:

- Pollen grain and ovules are developed in specialized structure

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flower. Seeds enclosed inside the fruits.

- Size varies from almost microscopic wolfia to tall tree Eucalyptus [more than 100m].



- Double fertilisation - Each pollen grain produce two male gametes. One gametes fuse with egg to form embryo, Syngamy and other gametes fuse with two polar nuclei to form endosperm, triple fission. Since fertilisation takes place twice so, it is called double fertilisation.

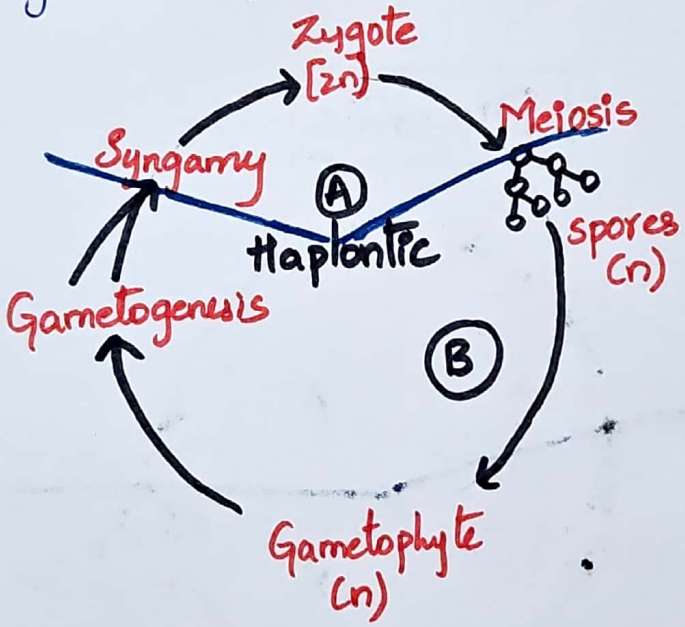
Monocotyledons	Dicotyledons
a) Single cotyledons.	a) Two cotyledons
b) Parallel venation	b) Reticulate venation
c) Fibrous root system	c) Tap root system
d) Closed vascular bundle	d) open vascular bundle
e) More number of vascular bundles	e) Less number of vascular bundles.
f) Banana, wheat, rice	f) Gram, mango, apple.

* **Alternation of generation:**

Different plant groups complete their life cycles in different

patterns. Angiosperms complete their life cycle in two phases - a diploid sporophytes and haploid gametophyte. The two follows each other. This phenomenon is called **alternation of generation.**

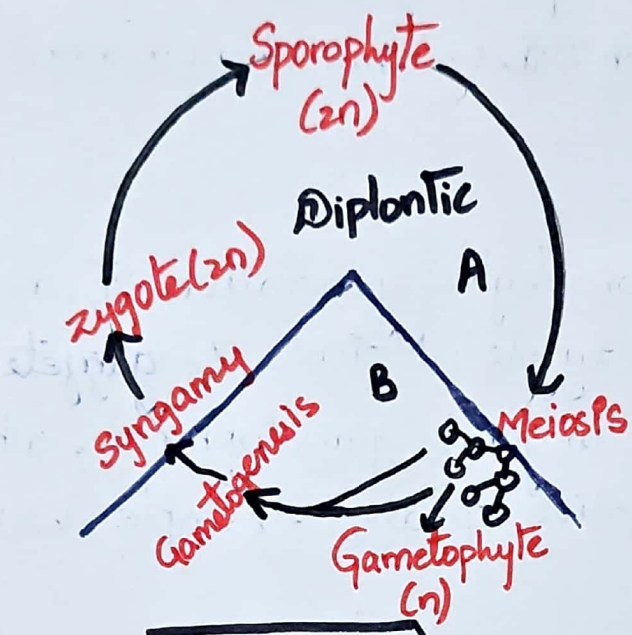
[a] Haplontic - Sporophytic generation is represented by only the one-celled zygote. Meiosis in zygote results into haploid spores to form gametophytes, which is the dominant vegetative phase. Example - Volvox, Spirogyra etc.



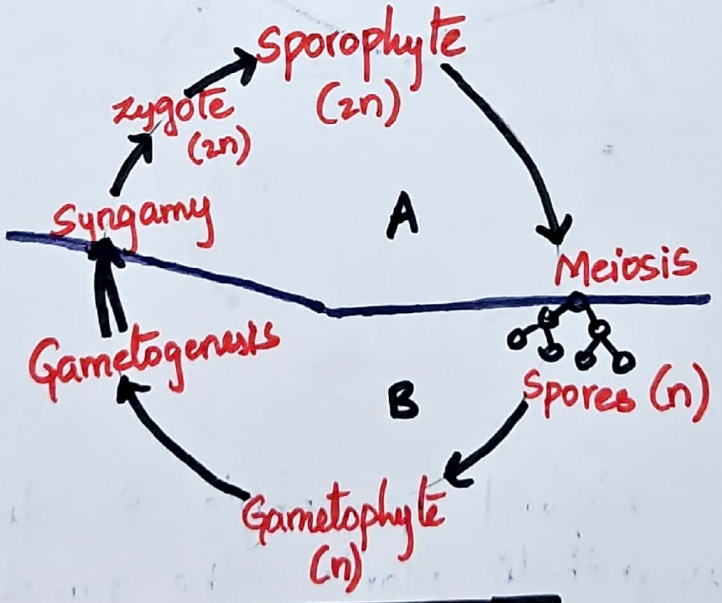
[b] Diplontic - Diploid sporophytes is dominant, interdep-endent, photosynthetic plants. The gametophyte is represented by single to few celled. All seed bearing plants fall under this category.

[c] Haplo-diplontic - Both phases are multicellular and intermediate condition is present. It is present in Bryophytes & pteridophytes.

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Diplontic



Haplo - Diplontic